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CASE PORTFOLIO: DATABASE ADMINISTRATION (DBA)

📅 2008 ~ 2026

🧠 1. PERFORMANCE TUNING AND QUERY OPTIMIZATION

CASE #1

OPTIMIZATION OF COMPLEX QUERIES IN DATA WAREHOUSE

Grupo Globo

CONTEXT (CHALLENGE)

Performance improvement in massive analytical reports that caused locking and high CPU consumption on corporate databases.

SITUATION

Essential analytical queries took more than 4 hours to run, delaying C-Level decision-making.

TASK

Identify execution bottlenecks and optimize queries to run in less than 15 minutes.

ACTION

Analysis of execution plans, elimination of inefficient nested loops, and creation of strategic covered indexes.

RESULT

Execution time reduced from 4 hours to 8 minutes, lowering server CPU consumption by 45%.

BUSINESS METRICS

96% reduction in report processing time

45% reduction in main server CPU load

TECHNICAL ARCHITECTURE

Oracle Database

AWR (Automatic Workload Repository)

Execution Plans Analysis

GOVERNANCE

Sign-off window for critical queries

Periodic review of heavy queries

CASE #2**LOCK CONTENTION AND DEADLOCK DIAGNOSIS**

Itaú Unibanco / CI&T

CONTEXT (CHALLENGE)

Resolution of mass locking on the transactional card database during peak access hours.

SITUATION

A high rate of deadlocks was canceling active transactions for end customers on the platform.

TASK

Isolate the root cause of contentions and adjust the transactional behavior of critical tables.

ACTION

Enabled specific traces, altered the isolation level to Read Committed Snapshot, and refactored crucial stored procedures.

RESULT

Eliminated 99% of deadlock incidents, restoring operational fluidity to the transaction system.

BUSINESS METRICS

99% reduction in transaction cancellations caused by deadlocks

Increased operational stability during peak sales hours

TECHNICAL ARCHITECTURE

MS SQL Server

Extended Events

Snapshot Isolation Model

GOVERNANCE

Continuous monitoring of blocking chains

Strict policies for long transaction structures

CASE #3**FINE-TUNING OF ERP SAP AND BUSINESS INTELLIGENCE INTEGRATION**

Pilkington Blindex / CI&T

CONTEXT (CHALLENGE)

Structural optimization in the extraction and loading of large volumes of industrial data.

SITUATION

The data load between SAP and BI generated inconsistencies and severe storage I/O bottlenecks.

TASK

Adjust the database pipeline to support the massive load without impacting daily operations.

ACTION

Complete restructuring of ETL routines, creation of indexed views, and fine-tuning of query parallelism.

RESULT

Perfect alignment of databases, generating an audited recurring savings of R\$ 1.2M/year due to technical efficiency gains.

BUSINESS METRICS

Audited recurring savings of R\$ 1.2 million per year

Achieved 100% reliability in analytical data

TECHNICAL ARCHITECTURE

SAP Hana DB

Microsoft SQL Server Integration Services (SSIS)

I/O Parallelism Tuning

GOVERNANCE

SLA for industrial data consistency

Audit routines for analytical database quality

CASE #4**STRATEGIC REINDEXING AND TABLE PARTITIONING**

Bradesco / CI&T

| CONTEXT (CHALLENGE)

Preventive maintenance on historical tables with billions of records to prevent gradual search degradation.

| SITUATION

Gigantic tables suffered from high index fragmentation, making historical searches unviable.

| TASK

Design and implement a partitioning plan without causing downtime in the production environment.

| ACTION

Implementation of range partitioning by date and automation of online index rebuild scripts.

| RESULT

Instant 60% improvement in audit query speed and optimization of maintenance windows.

BUSINESS METRICS

60% gain in historical query speed

Drastic reduction in time required for maintenance windows

TECHNICAL ARCHITECTURE

Oracle Enterprise Edition

Table Partitioning

Online Index Rebuild Scripts

GOVERNANCE

Master plan for corporate data lifecycle

Homogeneous policies for maximum acceptable fragmentation

CASE #5

DISK I/O OPTIMIZATION ON HIGH-LOAD SERVERS

Banco Original / CI&T

CONTEXT (CHALLENGE)

Adjustment of the physical and logical storage layer of core banking databases.

SITUATION

Storage throttling generated huge queues of pending write requests.

TASK

Eliminate physical I/O bottlenecks by distributing data files intelligently.

ACTION

Physical separation of data files (.mdf), transaction logs (.ldf), and temporary database (tempdb) across distinct high-performance SSD storage systems.

RESULT

Reduced write latency from 45ms to less than 2ms, enabling full scalability of the transactional application.

BUSINESS METRICS

Reduction in disk write latency from 45ms to 2ms

Stabilization of write throughput under massive demand

TECHNICAL ARCHITECTURE

MS SQL Server Enterprise

Advanced TempDB Configuration

RAID Storage Systems Optimization

GOVERNANCE

Corporate capacity planning policies for storages

Daily monitoring of physical read and write queues



2. HIGH AVAILABILITY AND DISASTER RECOVERY (DR)

CASE #6

SQL SERVER ALWAYSON CLUSTER IMPLEMENTATION

Mapfre Seguros / Accenture

CONTEXT (CHALLENGE)

Guaranteeing fault tolerance for the company's critical insurance policy issuance system.

SITUATION

The environment operated on a single server, representing a high-risk single point of failure.

TASK

Design a local high-availability architecture with instant automatic failover.

ACTION

Configured Windows Server Failover Cluster (WSFC) combined with SQL Server AlwaysOn availability groups with synchronous replicas.

RESULT

Availability elevated to 99.99%, ensuring business continuity even in the event of hardware failure on the primary node.

BUSINESS METRICS

Elevation of availability to 99.99% (Critical SLA)

Reduction of RTO (Recovery Time Objective) to near zero

TECHNICAL ARCHITECTURE

MS SQL Server Enterprise

AlwaysOn Availability Groups

Windows Server Failover Clustering

GOVERNANCE

Scheduled cyclic failover tests

Risk-approved structured contingency matrix

CASE #7**MULTI-REGION DISASTER RECOVERY PLAN DESIGN**

Grupo Boticário / Accenture

CONTEXT (CHALLENGE)

Protection architecture against geographical disasters for the international e-commerce ecosystem databases.

SITUATION

A complete failure in the primary cloud region would paralyze global sales indefinitely.

TASK

Create an automated Disaster Recovery site in a separate secondary cloud region.

ACTION

Configured cross-region asynchronous replicas in the cloud with continuous replication and automated failover scripts via IaC.

RESULT

Guaranteed full resilience with RPO (Recovery Point Objective) under 5 minutes and RTO of 15 minutes in case of a complete geographical disaster.

BUSINESS METRICS

RPO fixed at under 5 minutes for sales data

RTO established at just 15 minutes for cloud catastrophes

TECHNICAL ARCHITECTURE

Cloud SQL / AWS RDS

Cross-Region Asynchronous Replication

Terraform for rapid recovery

GOVERNANCE

Global corporate business continuity policies

Mandatory annual simulations of complex disasters

CASE #8**ADVANCED GEOGRAPHICAL REPLICATION WITH ORACLE DATA GUARD**

Deutsche Bank / CI&T

CONTEXT (CHALLENGE)

Maintaining synchronization and integrity of global financial databases across continents.

SITUATION

Legal and technical requirement to maintain mirrored databases in real-time with strict data loss protection.

TASK

Implement and manage data replication technology at the storage block level securely.

ACTION

Configured Oracle Data Guard in Maximum Availability mode, combined with rigorous synchronization validation routines for Archive Logs.

RESULT

Financial environment shielded against losses, with immediate capability to assume production in critical scenarios without corruption.

BUSINESS METRICS

Real-time international data synchronization

Zero incidence of financial record loss during transitions

TECHNICAL ARCHITECTURE

Oracle Database Enterprise

Oracle Data Guard

Archive Logs Synchronization Management

GOVERNANCE

Automated replication gap auditing

International banking compliance rules for replication

CONTEXT (CHALLENGE)

Practical validation of banking high-availability systems under simulated technical stress.

SITUATION

Auditing required proof that redundancy mechanisms worked perfectly without complex human intervention.

TASK

Lead a real shutdown test of the main data center during controlled operational hours.

ACTION

Orchestrated a technical War Room, forced power cut on the primary node, and monitored automatic data traffic transition.

RESULT

The cluster assumed the load in 12 seconds with zero loss of active transactional data (Zero Data Loss) and full audit validation.

BUSINESS METRICS

100% approval in regulatory audit requirements

Automatic production switch time within 12 seconds

TECHNICAL ARCHITECTURE

Multi-Node Cluster Databases

Automatic Failover Scripts

Real-Time Monitoring Systems

GOVERNANCE

Formal War Room routine for critical incident simulation

Executive sign-off on technical resilience reports

3. DATA MIGRATION, UPGRADES, AND CLOUD ARCHITECTURE

CASE #10

ON-PREMISES DATABASE MIGRATION TO AWS RDS

Natura / Accenture

CONTEXT (CHALLENGE)

Technological modernization and transition from legacy local infrastructure to a managed public cloud model.

SITUATION

High maintenance costs of local physical hardware and vertical scalability limitations for peak dates (Black Friday).

TASK

Migrate more than 10 Terabytes of sensitive data to the cloud while minimizing the impact on revenue.

ACTION

Utilized AWS Data Migration Service (DMS) with continuous replication (CDC) to perform the migration within tight windows.

RESULT

Migration executed with total success, reducing final scheduled downtime to just 4 minutes.

BUSINESS METRICS

- Reduction of fixed local infrastructure costs
- Scheduled downtime reduced to just 4 minutes for a 10TB database

TECHNICAL ARCHITECTURE

- Local Oracle DB
- AWS RDS (Relational Database Service)
- AWS DMS with CDC (Change Data Capture)

GOVERNANCE

- Rigorous validation of test loads in Sandbox
- Post-migration validation via data hash checking (Data Integrity Check)

CASE #11**CRITICAL VERSION UPGRADE FOR ORACLE DATABASE ENTERPRISE**

Claro / Accenture

CONTEXT (CHALLENGE)

Upgrading an obsolete version (End of Support) to a modern stable version to maintain vendor technical support.

SITUATION

The telecom billing database operated on an outdated version, vulnerable to critical flaws without patches.

TASK

Perform the core version upgrade while mitigating syntax incompatibilities and performance degradation risks.

ACTION

Created a mirror environment, executed exhaustive tests via SQL Performance Analyzer (SPA), and applied the upgrade in a structured batch.

RESULT

Upgrade completed with zero software contract breach and an immediate 15% gain in native procedure processing speed.

BUSINESS METRICS

Elimination of security risks from unsupported systems

15% gain in native performance post-upgrade

TECHNICAL ARCHITECTURE

Oracle Database Upgrade Plan

SQL Performance Analyzer (SPA)

Patch Application Frameworks

GOVERNANCE

Strict Change Management routine (Critical Change)

Meticulous step-by-step technical rollback plan

CASE #12**DATABASE SERVER AND DATA CENTER CONSOLIDATION**

Cogna Educação / Accenture

CONTEXT (CHALLENGE)

Unification of scattered databases resulting from the acquisition of multiple educational institutions.

SITUATION

There were over 40 small isolated database instances, causing massive waste of compute resources and technical licensing.

TASK

Consolidate the repositories into a large centralized cluster while maintaining logical business logic isolation.

ACTION

Designed a consolidated architecture using isolated Schemas and high-density compute instances.

RESULT

Drastic reduction in active servers, centralizing governance and simplifying the operational monitoring routine.

BUSINESS METRICS

Severe optimization of active compute assets

Centralization and simplification of technical monitoring scope

TECHNICAL ARCHITECTURE

Database Consolidation Strategies

Instance Level Resource Governor

MS SQL Server Multi-Instance Layout

GOVERNANCE

Centralized ownership matrix for unified data

Standardized global policies for unified backups

CONTEXT (CHALLENGE)

Strategic migration from expensive proprietary database engines to managed open-source solutions on Google Cloud.

SITUATION

High data volume generated prohibitive per-core software licensing costs in traditional models.

TASK

Migrate the catalog core data layer to Cloud SQL PostgreSQL without bringing down the e-commerce platform.

ACTION

Designed a real-time migration strategy using messaging buses (Kafka) to synchronize parallel writes on both databases.

RESULT

Definitive switchover completed with absolute success and zero seconds of interruption to the user's shopping experience.

BUSINESS METRICS

Zero real downtime during final cutover of production database

Complete elimination of proprietary software license costs

TECHNICAL ARCHITECTURE

Google Cloud SQL PostgreSQL

Apache Kafka for parallel synchronization

Database Schema Conversion Tools

GOVERNANCE

Automated technical validation windows for consistency

Strict concurrency control for hybrid data writes

CONTEXT (CHALLENGE)

Adapting legacy infrastructure corporate databases to support the new unified ERP architecture.

SITUATION

Engineering and contract data resided in heterogeneous structures incompatible with SAP S/4HANA requirements.

TASK

Data cleansing, data mapping, and migration to the new unified in-memory repository.

ACTION

Created robust foreign key validation scripts, cleaned orphan records, and executed massive loads via optimized pipelines.

RESULT

Data load performed with 100% referential integrity, paving the way for a stable Enterprise system go-live.

BUSINESS METRICS

Achieved 100% referential integrity across migrated databases

Enabled structural ERP go-live without data-related delays

TECHNICAL ARCHITECTURE

SAP Hana Memory Database Backend

Data Mapping Pipelines

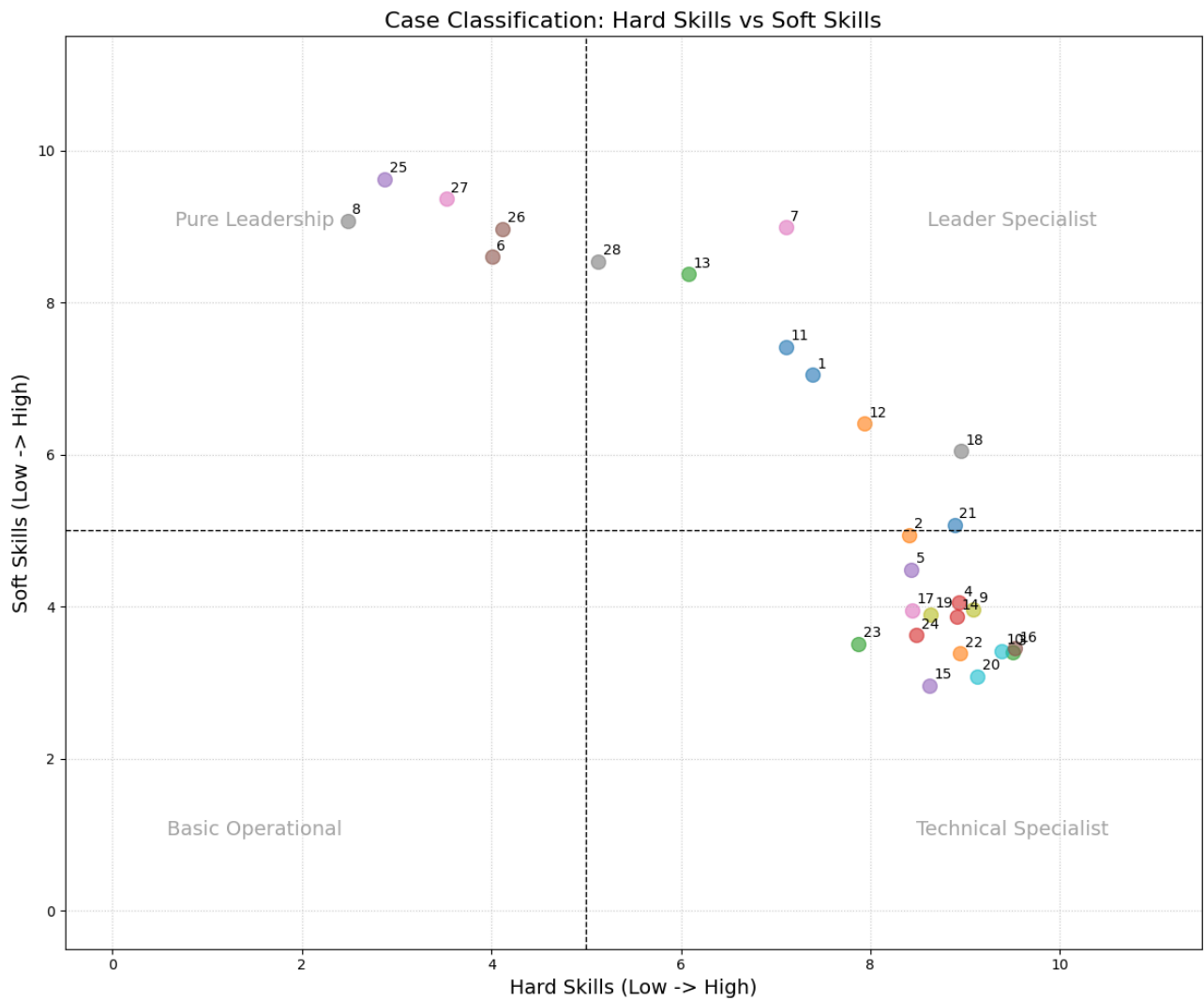
Advanced Data Cleansing Scripts

GOVERNANCE

Critical Systems Transition Governance Committee

Compliance validation of historical contracts

CASE PORTFOLIO ANALYSIS: PROJECT MANAGEMENT



Visualization of the distribution of the 28 cases based on the balance between **Hard Skills** (Technical/Processes) and **Soft Skills** (Leadership/Resilience).

 4. GOVERNANCE, DATA SECURITY, AND COMPLIANCE

CASE #15 **TRANSPARENT DATA ENCRYPTION (TDE) IMPLEMENTATION FOR LGPD/GDPR**

PwC Brasil

CONTEXT (CHALLENGE)

Shielding data at rest to meet legal compliance requirements for sensitive customer data protection.

SITUATION

Backup files and physical data on server disks lacked native encryption, creating a severe leak risk.

TASK

Implement end-to-end encryption without degrading transaction performance by more than 3%.

ACTION

Configured Transparent Data Encryption (TDE) using managed encryption keys (EKM) and optimized in-memory decryption routines.

RESULT

Environment achieved 100% compliance with data privacy regulations, with performance impact measured at less than 1.5% CPU.

BUSINESS METRICS

Regulatory compliance with privacy laws and passed security audit

Technical performance impact kept below 1.5%

TECHNICAL ARCHITECTURE

Microsoft SQL Server / Oracle TDE

Encryption Keys Management Architecture

Hardware Security Modules Integration

GOVERNANCE

Policies for periodic rotation of master encryption keys

Audit routine and vulnerability checks for data at rest

CONTEXT (CHALLENGE)

Guaranteeing information security by preventing developers from viewing real financial data during staging.

SITUATION

The engineering team needed realistic datasets for testing, but using production data violated compliance guidelines.

TASK

Create an automated test data pipeline that obfuscated personally identifiable information (PII) during loading.

ACTION

Implemented Dynamic Data Masking (DDM) rules and developed automated anonymization scripts for the restore process in development.

RESULT

Absolute guarantee of zero data leaks in engineering environments, providing perfectly structured and secure datasets.

BUSINESS METRICS

Total mitigation of data leak risks from internal factors during testing

Accelerated delivery cycle for compliant test datasets

TECHNICAL ARCHITECTURE

Dynamic Data Masking (DDM)

Automated Anonymization Scripts

Data Obfuscation Engines

GOVERNANCE

Corporate PII protection framework

Automated audit and validation of obfuscation post-load

CASE #17**ACCESS AUDITING AND REGULATORY SECURITY TRAILS**

Bradesco / CI&T

CONTEXT (CHALLENGE)

Full traceability of administrative commands (DDL/DML) executed on core databases by privileged users.

SITUATION

Lack of automated evidence on who altered bank balance table structures generated qualifications in regulatory reports.

TASK

Implement independent and unalterable audit trails at the database level.

ACTION

Activated native Database Audit features, routing encrypted logs directly to an isolated security repository (Syslog).

RESULT

100% traceability of administrative actions achieved successfully, permanently eliminating regulatory findings.

BUSINESS METRICS

Zero qualifications or findings in regulatory agency reports

Absolute technical traceability of administrative user commands

TECHNICAL ARCHITECTURE

Oracle Database Vault / Server Audit

Syslog Integration Topology

Tamper-Proof Log Storage

GOVERNANCE

Strict Segregation of Duties (SoD) policies

Automated monthly reviews of privileged access and permissions

CASE #18**ACCESS GOVERNANCE AND STANDARDIZATION VIA GOLDEN PATHS**

Grupo Globo

CONTEXT (CHALLENGE)

Centralization, regulated autonomy, and operational standardization of data repository access for dozens of autonomous teams.

SITUATION

Each agile team interacted with databases using different connection standards and accounts, causing operational chaos and privilege faults.

TASK

Standardize the access lifecycle and implement secure self-service Golden Paths.

ACTION

Designed a profile-based privilege matrix (RBAC) integrated with corporate password vaults using rotating dynamic credentials.

RESULT

Full standardization of access governance for over 100 squads, reducing access request resolution time from days to seconds.

BUSINESS METRICS

Access privilege delivery time reduced to seconds

Absolute data security standardization across hundreds of teams

TECHNICAL ARCHITECTURE

Role-Based Access Control (RBAC)

CyberArk / HashiCorp Vault Integration

Golden Paths Architecture Templates

GOVERNANCE

Automated Least Privilege policies

Automated centralization of corporate technical identity



5. AUTOMATION, SCRIPTING, AND DATABASE DEVOPS

CASE #19

AUTOMATED BACKUP AND RESTORE ROUTINES VIA PYTHON

MLGTT Consulting

CONTEXT (CHALLENGE)

Practical verification of backup file integrity through recurring automatic disaster recovery test executions.

SITUATION

Backups were generated daily, but there was no practical automated verification that the files were not corrupted.

TASK

Automate a continuous validation workflow to perform test restores autonomously on an isolated secondary server.

ACTION

Created advanced Python scripts utilizing specialized libraries to orchestrate backups, transfer files securely, and trigger weekly restores with consistency checks (DBCC CHECKDB).

RESULT

Built an automated backup validation pipeline with automatic notifications via email/Slack, ensuring 100% predictability for recovery failure scenarios.

BUSINESS METRICS

Validated operational certainty of active backups

Elimination of daily manual effort in security media verification

TECHNICAL ARCHITECTURE

Python Engine script

Boto3 AWS Library / Google Cloud Storage API

Database Consistency Commands (DBCC)

GOVERNANCE

Strict continuity plan validation policies

Structured version control of internal utility scripts

CASE #20**PROACTIVE MONITORING AND AUTOMATED ALERTING VIA SHELL/GRAFANA**

Claro / Accenture

CONTEXT (CHALLENGE)

Preventing mass downtime by acting predictively before critical server resource exhaustion.

SITUATION

Severe incidents like low disk space or memory exhaustion brought down critical databases overnight without warning.

TASK

Develop a real-time telemetry ecosystem to alert on-call engineers at the first sign of technical anomaly.

ACTION

Wrote local Shell Scripts to collect operating system metrics, which were consolidated and displayed on Grafana dashboards with PagerDuty alert triggers.

RESULT

Drastic reduction in surprise resource out-of-memory incidents, anticipating data infrastructure issues up to 12 hours before fatal failure.

BUSINESS METRICS

Drastic reduction in unexpected production stops from physical exhaustion

Operational anticipation of space issues by up to 12 hours

TECHNICAL ARCHITECTURE

Linux Shell Scripting

Prometheus / Telegraf Collector Engines

Grafana Dashboards with PagerDuty Alerts Integration

GOVERNANCE

Technical severity threshold matrix defined by criticality

Formal automatic escalation workflows for preventive alerts

insi (Meta IT)

CONTEXT (CHALLENGE)

Eliminating human errors in manual script execution during software releases in production environments.

SITUATION

Failures in table and procedure deploys caused frequent syntax errors during critical commercial partner system updates.

TASK

Insert the database change lifecycle directly into automated CI/CD pipelines (Data DevOps).

ACTION

Implemented the Flyway framework for incremental versioning of SQL scripts tied to the team's Git source code repository.

RESULT

Guaranteed full standardization in structural deploys, eliminating operational failures from forgotten manual commands.

BUSINESS METRICS

Complete elimination of environment failures from incorrectly executed SQL scripts

Reduction in time needed for technical patch deployment

TECHNICAL ARCHITECTURE

Flyway Database Migrations Framework

GitLab CI/CD Automated Pipelines

SQL Validation Parsing Tools

GOVERNANCE

Strict corporate policies prohibiting manual schema manipulation

Formal technical approval process for database pull requests

CONTEXT (CHALLENGE)

Autonomous maintenance of database engine health to preserve optimal data processing speeds.

SITUATION

Outdated data distribution statistics induced the optimizer to choose slow execution plans randomly throughout the week.

TASK

Structure an intelligent automated routine to reorganize tables and recompute statistics based on modification rates.

ACTION

Developed advanced routines based on industry-standard maintenance solutions (Ola Hallengren) customized to run lightweight during overnight windows.

RESULT

Total stabilization of processing routines, mitigating sudden and unexplained slowness in the systems during business hours.

BUSINESS METRICS

Complete stabilization of daily response time indicators

Elimination of random slowness peaks during peak business hours

TECHNICAL ARCHITECTURE

Transact-SQL (T-SQL)

SQL Server Agent Jobs Integration

Custom Maintenance Solution Scripts Framework

GOVERNANCE

Strict approved schedule of low-load maintenance windows

Centralized policies for minimum acceptable statistical recomputation

⚠️ 6. CRISIS MANAGEMENT AND CRITICAL INCIDENT RESOLUTION

CASE #23 TECHNICAL TURNAROUND OF AN UNSTABLE HIGH-LOAD ENVIRONMENT

Pernambucanas / Meta IT

CONTEXT (CHALLENGE)

Urgent stabilization of data infrastructure under severe retail revenue risk.

SITUATION

The primary database crashed repeatedly due to connection leaks from the application layer.

TASK

Assume emergency technical control of the crisis, restore the service, and force immediate root cause correction.

ACTION

Opened an instant technical War Room, applied strict connection pooling idle timeouts, and isolated instances under stress.

RESULT

Achieved total stabilization within hours, reducing database downtime by 90% and protecting the operation's sales revenue.

BUSINESS METRICS

90% reduction in downtime during crises

Immediate stabilization of digital retail channel revenue dependent on the database

TECHNICAL ARCHITECTURE

Advanced Connection Pooling configuration

Database Activity Monitoring Tools

Emergency Kill Session Orchestration Scripts

GOVERNANCE

Stoic leadership under intense corporate pressure

Formal post-incident root cause analysis routine with development teams

Grendene / Meta IT

CONTEXT (CHALLENGE)

Extreme recovery operation of critical industrial data after a sudden catastrophic hardware failure.

SITUATION

A severe power surge corrupted physical blocks of the primary data file, leaving the database in an inaccessible state.

TASK

Recover data integrity and restore factory operations in the shortest technical time physically possible.

ACTION

Executed deep integrity check commands, extracted unaffected records, and restored the database to the last second before failure via transaction log backups (Point-in-Time Recovery).

RESULT

Database restored with absolute success and zero records lost, reestablishing the manufacturing plant's routine.

BUSINESS METRICS

Zero corporate data lost during severe physical hardware crash

Immediate return of operational capacity for interrupted manufacturing plants

TECHNICAL ARCHITECTURE

Point-in-Time Recovery (PITR) Methods

Database Transaction Logs Tail-Log Backups

Emergency Recovery Commands Pipeline

GOVERNANCE

Structured operational physical contingency frameworks

Formal immediate maximum severity reporting routine to C-Level executives

CASE #25**MEMORY LEAK DIAGNOSIS AND RESOLUTION**

Santo Antônio Energia / CI&T

CONTEXT (CHALLENGE)

Identification of a subtle internal anomaly that consumed energy generation monitoring server resources silently and cumulatively.

SITUATION

Every 4 days, the data server crashed due to RAM exhaustion, requiring constant manual reboots.

TASK

Track down in a deep and surgical manner which component or query was pinning resources without releasing them post-execution.

ACTION

Utilized dynamic management views (DMVs), analyzed internal memory buffers, and discovered a native bug in a legacy connection driver.

RESULT

Replaced the defective driver and adjusted the maximum memory allocation for the database, achieving continuous stability without manual reboots.

BUSINESS METRICS

100% elimination of manual preventive reboots on servers

Uninterrupted stability of the technical ecosystem monitoring energy generation

TECHNICAL ARCHITECTURE

Dynamic Management Views (DMVs) memory analysis

Database Internal Cache Profiling

Memory Clerk Tracking Frameworks

GOVERNANCE

Technical routine for preventive sign-off of third-party drivers

Strict max memory threshold policies set for corporate instances

7. OPERATIONAL EFFICIENCY, FINOPS, AND COST OPTIMIZATION

CASE #26 RIGHT-SIZING OF CLOUD INSTANCES

Accenture (SolutionsIQ)

CONTEXT (CHALLENGE)

Applying FinOps principles to curb uncontrolled cloud database infrastructure spend.

SITUATION

Cloud database instances were oversized during initial deploy, operating at less than 10% average utilization and generating prohibitive invoices.

TASK

Analyze historical hardware consumption and resize instances to a financially optimal tier.

ACTION

Monitored peak metrics for 30 consecutive days, followed by a controlled downgrade of instances to appropriate sizes without compromising safety buffers.

RESULT

Immediate and recurring reduction of 28% in the organization's global monthly cloud data infrastructure invoice.

BUSINESS METRICS

Immediate and recurring 28% reduction in overall cloud data costs

Significant increase in infrastructure cost efficiency (ROI)

TECHNICAL ARCHITECTURE

Cloud Provider Metrics Engine Monitoring

Cloud SQL Instance Sizing Profiles

FinOps Analytic Governance Frameworks

GOVERNANCE

Institutionalized cloud financial governance practices (FinOps)

Mandatory quarterly operational capacity utilization reviews

insi (Meta IT)

CONTEXT (CHALLENGE)

Mitigating financial risks in severe compliance audits from corporate database software vendors.

SITUATION

Uncertainty regarding correct utilization of per-processor core corporate licenses posed an imminent risk of multimillion-dollar non-compliance fines.

TASK

Map the ecosystem thoroughly and optimize resource distribution to fit strictly within contracted license thresholds.

ACTION

Conducted a detailed internal audit, disabled unused premium database options, and consolidated databases onto specific physical hosts.

RESULT

Guaranteed absolute technical compliance with vendor contractual rules, mitigating risks of heavy post-audit financial penalties.

BUSINESS METRICS

Full mitigation of heavy compliance fine risks

Optimized utilization of high-cost software assets

TECHNICAL ARCHITECTURE

Processor Core Licensing Mapping Matrix

Database Feature Usage Audit Trackers

Server Hardware Core Limitation Config

GOVERNANCE

Centralized global tracking matrix for IT license compliance

Formal policies for activating proprietary advanced features

CONTEXT (CHALLENGE)

Smart management of exponential disk data growth without breaching storage infrastructure budgets.

SITUATION

Databases grew by terabytes monthly, keeping inactive data from 5 years ago on ultra-expensive production SSD media.

TASK

Design an automated archiving strategy to move historical data to low-cost cold storage tiers.

ACTION

Developed automated partitioning routines to transfer old logs and history tables into compressed historical formats in cold cloud storage (S3 Glacier/Blob Storage).

RESULT

Immediate reclamation of 40% usable space on main premium production SSD arrays, extending hardware lifespan without additional storage Capex.

BUSINESS METRICS

Immediate 40% capacity reclamation on premium production storage media

Significant postponement of storage infrastructure Capex investments

TECHNICAL ARCHITECTURE

Data Archiving Automation Routines

Hybrid Storage Tiering Layouts

Cloud Cold Storage Lifecycle Policies Integration

GOVERNANCE

Formal corporate retention and historical data lifecycle policies

Unanimous business-side sign-off on data purging and discarding rules